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[Working Title] Generation With Fine-grained Citations

AVG Spotlight Presentation 1

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Problem

Given a document A (which may be generated) with a citation c , find the passage p in the cited document B that is most relevant to A .

If A is generated, p should ideally be the passage in B that *caused* the LLM to generate the statement that cites B (this problem is introduced as *context attribution* in [1]).

Related Work

ScholarCopilot [7]

- RAG with dynamic retrieval during generation via the generation of retrieval tokens
- focused on academic writing, currently introduction and related work sections

ContextCite [1]

- method for scoring candidate context passages for *contributive attribution* ('*is this source the reason this statement was generated?*')
- determines how the prob. of the generated statement changes in a specific source is excluded
- evaluated on Wikipedia QA and news summarization datasets



Related Work, continued

ReasonIR [3]

- model+dataset for reasoning-intensive long context retrieval
- they also propose a method for generating synthetic training data
- dataset based on MS MARCO, Natural Questions + HotpotQA



Method - Dataset

- citing document *A*: Fastformer [8]
 - citation context window size: 128 tokens
 - context window centered at the citation
- cited document *B*: Transformer [5]
 - sections / subsections split into chunks
 - chunk size: 512 tokens
- this yields 4 citations (2x in introduction, 1x in methods and in experiments)



Method - Model

Retriever: ReasonIR-8B [3]

- LLaMA3.1-8B fine-tuned on the ReasonIR dataset

Reranker: Qwen2.5-7B-Instruct [4, 9]

- additionally, the reranking score for each passage is averaged over m LLM calls (this implements a simple version of batched self-consistency [2], see also [6])
- I currently use $m_{\text{retrieval}} = m_{\text{reranking}} = 10$

Final Score

- interpolation of both ranking scores:
$$s_{\text{final}} = (1 - \lambda)s_{\text{retrieval}} + \lambda s_{\text{reranking}}$$
 (I currently use $\lambda = 0.5$)

All scores are currently obtained by comparing the query to every candidate document individually, this will be changed in the near future!

Current Results (Good Example)

section-chunk	$S_{retrieval}$	$S_{reranking}$	S_{final}
Conclusion-0	0.627	0.8	0.71
abstract-0	0.547	0.25	0.4
Background-0	0.59	0.14	0.37
Introduction-0	0.594	0.06	0.33
Model Architecture-0	0.589	0	0.29

Figure: Ranking scores for query-0. Ranked are passages extracted from the Transformer paper.

query-0 (first citation in the Fastformer paper)

[...] Transformer models and can meanwhile achieve comparable or even better long text modeling performance. Introduction Transformer (<CIT-MASK>) and their variants have achieved great success in many fields. For example, Transformer is the backbone architecture of many state-of-the-art pre-trained language models in NLP, such as [...]



Current Results (Bad Example)

section-chunk	$S_{retrieval}$	$S_{reranking}$	S_{final}
section-chunk	$S_{retrieval}$	$S_{reranking}$	S_{final}
section-chunk	$S_{retrieval}$	$S_{reranking}$	S_{final}
section-chunk	$S_{retrieval}$	$S_{reranking}$	S_{final}
section-chunk	$S_{retrieval}$	$S_{reranking}$	S_{final}
section-chunk	$S_{retrieval}$	$S_{reranking}$	S_{final}

Figure: Ranking scores for query-.

query-

[...] [...]



Outlook

TODO

- [1] Cohen-Wang, Benjamin, Harshay Shah, Kristian Georgiev & Aleksander Madry (2024). *ContextCite: Attributing Model Generation to Context*.
<https://arxiv.org/abs/2409.00729>.
- [2] Korikov, Anton, Pan Du, Scott Sanner & Navid Rekabsaz (2025). *Batched Self-Consistency Improves LLM Relevance Assessment and Ranking*.
<https://arxiv.org/abs/2505.12570>.
- [3] Shao, Rulin, Rui Qiao, Varsha Kishore, Niklas Muennighoff, Xi Victoria Lin, Daniela Rus, Bryan Kian Hsiang Low, Sewon Min, Wen-tau Yih, Pang Wei Koh & Luke Zettlemoyer (2025). *ReasonIR: Training Retrievers for Reasoning Tasks*.
<https://arxiv.org/abs/2504.20595>.
- [4] Team, Qwen (September 2024). *Qwen2.5: A Party of Foundation Models*.
<https://qwenlm.github.io/blog/qwen2.5/>.
- [5] Vaswani, Ashish, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser & Illia Polosukhin (2023). *Attention Is All You Need*.
<https://arxiv.org/abs/1706.03762>.
- [6] Wang, Xuezhi, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery & Denny Zhou (2023). *Self-Consistency Improves Chain of Thought Reasoning in Language Models*. <https://arxiv.org/abs/2203.11171>.

- [7] Wang, Yubo, Xueguang Ma, Ping Nie, Huaye Zeng, Zhiheng Lyu, Yuxuan Zhang, Benjamin Schneider, Yi Lu, Xiang Yue & Wenhua Chen (2025). *ScholarCopilot: Training Large Language Models for Academic Writing with Accurate Citations*.
<https://arxiv.org/abs/2504.00824>.
- [8] Wu, Chuhan, Fangzhao Wu, Tao Qi, Yongfeng Huang & Xing Xie (2021). *Fastformer: Additive Attention Can Be All You Need*. <https://arxiv.org/abs/2108.09084>.
- [9] Yang, An, Baosong Yang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Zhou, Chengpeng Li, Chengyuan Li, Dayiheng Liu, Fei Huang, Guanting Dong, Haoran Wei, Huan Lin, Jialong Tang, Jialin Wang, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Ma, Jin Xu, Jingren Zhou, Jinze Bai, Jinzheng He, Junyang Lin, Kai Dang, Keming Lu, Keqin Chen, Kexin Yang, Mei Li, Mingfeng Xue, Na Ni, Pei Zhang, Peng Wang, Ru Peng, Rui Men, Ruize Gao, Runji Lin, Shijie Wang, Shuai Bai, Sinan Tan, Tianhang Zhu, Tianhao Li, Tianyu Liu, Wenbin Ge, Xiaodong Deng, Xiaohuan Zhou, Xingzhang Ren, Xinyu Zhang, Xipin Wei, Xuancheng Ren, Yang Fan, Yang Yao, Yichang Zhang, Yu Wan, Yunfei Chu, Yeqiong Liu, Zeyu Cui, Zhenru Zhang & Zhihao Fan. 2024. Qwen2 Technical Report. In *arXiv preprint arXiv:2407.10671*.